System software and modes of processing

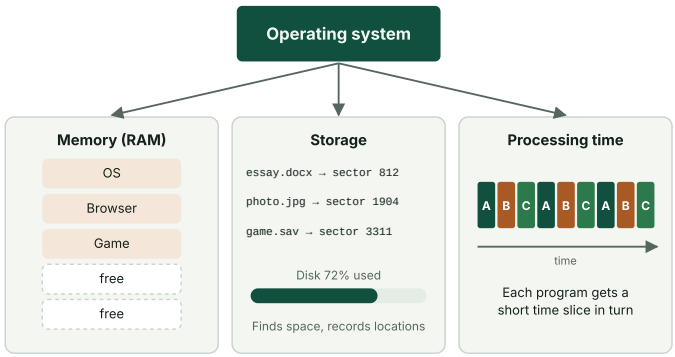
L06
FACT FILE

Specification: students should be able to

- Distinguish between system software and application software
- Describe the functions of system software, referring to allocating memory, storage and processing time
- Describe the boot-up sequence
- Describe the following modes of processing: real-time, batch and multi-user
- Choose and justify a mode of processing for a given application

CCEA fact file: Software, Software 2

Key words: software, system software, application software, operating system, memory management, processing time, time slice, boot-up sequence, real-time processing, batch processing, multi-user system, round robin, master file, transaction file, packaged software, tailored software, device driver



The operating system allocating memory, storage and processing time

System software and application software

Software is the set of instructions that tells the hardware what to do. It is split into two types. System software runs the computer itself. It acts as the link between the hardware and the programs the user wants to run. The most important piece of system software is the operating system, such as Windows, macOS, Android or iOS, which manages every other program. Utility programs and device drivers are also system software. Application software lets the user carry out a particular task, such as writing an essay, editing a photo or browsing the web. Applications can be packaged (off-the-shelf), which means they are made for many users and are cheap and ready to use, or tailored (bespoke), which means they are written for one organisation. Tailored software fits the organisation's needs exactly, but it costs more and takes time to develop, because one customer pays for all the design, programming and testing.

How the operating system manages resources

The operating system shares out the computer's resources. First, it allocates memory. A program or file must be copied from storage into RAM before the processor can use it, so the OS finds free space in RAM, gives it to the program and keeps programs apart so they do not overwrite each other. When the program closes, the space is freed. Second, it allocates storage. The OS saves files on the storage drive, records where each one is, frees the space when files are deleted and controls who can open them. Third, it allocates processing time. The processor can only do one thing at a time on each core, so the OS splits its time into tiny time slices and hands them to each task in turn. Tasks with a higher priority receive more slices. The switching is so quick that several programs appear to run at once. The OS also starts the computer: when it is switched on, instructions held in ROM check the hardware and load the operating system into RAM.

Real-time and batch processing

In real-time processing, data is processed as soon as it is entered and the output is ready quickly enough to affect the next input. Booking systems need this. When someone buys a ticket for a concert, the seat must be marked as sold straight away, or another person could buy the same seat. Air traffic control is even more critical, because an aircraft's position on screen must always be up to date. In batch processing, data is collected over a period, such as a week or a month, and then processed in one go without people being involved. It suits large amounts of similar data where an instant result is not needed. An electricity company keeps a master file of customer details. New meter readings go into a temporary transaction file. Overnight, each reading is matched to its customer using the account number, the bill is calculated, all the bills are produced together, and the master file is updated for next time.

Multi-user systems

A multi-user system lets many people, each at a different computer or terminal, share the processor of one powerful computer at the same time. It is also called a time-sharing system. The operating system gives each user a very short time slice of processor time in turn. When it has served the last user it goes back to the first, which is known as the round robin method. Because the processor works so quickly, each person feels as if they have the whole computer to themselves. For example, if four users each get a 20-millisecond slice, every user gets a turn every 80 milliseconds, which is twelve and a half times every second. Multi-user is not the same as real-time. Real-time describes how quickly data is processed, while multi-user describes how one processor is shared between many people. A system can be both, such as an online booking system used by many customers at once.

Key terms from the mark schemes

System software
Software that runs and manages the computer hardware and acts as the interface between the hardware and application programs, e.g. the operating system, utilities and device drivers.

Application software
Software designed to let the user carry out a specific task, such as word processing, spreadsheets or web browsing.

Operating system
System software that manages the computer's resources – allocating memory, storage and processing time – and lets the hardware and application software work together.
Credited: U1 2024 Q10, 2025 Q4

Boot-up sequence
The steps carried out when a computer is switched on: instructions in ROM check the hardware and load the operating system from storage into RAM.
Credited: U1 2025 Q4

Real-time processing
Each input is processed the moment it is entered, so the output is produced in time to affect the next input, e.g. ticket booking or air traffic control.
Credited: U1 2022 Q1, 2023 Q5

Batch processing
Similar data is collected into batches over a period and processed together later, without human involvement, often overnight, e.g. bills and payroll.
Credited: U1 2023 Q12

Multi-user system
A system in which many users at different terminals share the processor of one powerful computer, each given a time slice in turn (round robin).
Credited: time-sharing system; U1 2023 Q5

Time slice
A short period of processor time given to a task or user before the processor moves on to the next one.
Credited: U1 2019 Q5, 2023 Q5

Master file
A permanent or semi-permanent file holding main records (e.g. customer details and reading history) that is updated when batch processing takes place.

Transaction file
A temporary file holding the transactions collected for a batch (e.g. meter readings); it is used to update the master file.

Packaged software
Off-the-shelf software made for many users with common needs; low cost and ready to use, e.g. Microsoft Word.

Tailored software
Bespoke software developed for one organisation's specific needs; usually expensive because one customer bears the whole cost.

Exam focus

Questions on this lesson often use command words such as describe and distinguish, and they nearly always want an example. Real-time processing appeared in 2022 Q1 and was compared with multi-user processing in 2023 Q5(c). Strong answers said real-time processes each input immediately so the output affects the next input, and that multi-user shares one processor between many users using time slices. 2023 Q5(b) asked which resources the operating system manages: name memory, storage and processing time, and say what the OS does with each. Batch processing came up in 2023 Q12(a); mention collecting data over time, processing it together without human input, and a suitable example such as bills or payroll. In 2025 Q4 the Chief Examiner reported that many pupils could not describe the boot-up sequence, so learn the order: ROM instructions run, hardware is checked, the OS is loaded into RAM. Key-term matching (2024 Q10) also tested definitions from this lesson.

Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2024 Q10	[8]	Key term matching
Unit 1 2023 Q12	[4]	Batch processing, backups, freeing disk space
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. Describe the difference between system software and application software. [2]
2. Classify each software example as either system software (S) or application software (A): Windows 11 operating system; Microsoft Excel spreadsheet; patient record database; disk defragmenter software; device driver; Google Chrome web browser. [6]
3. Describe two differences between packaged software and tailored software. [4]

Answers: 1. System software runs / manages the computer hardware, e.g. the operating system (1) / application software lets the user carry out a specific task, e.g. word processing (1). 2. Windows 11 – S (1) / Excel – A (1) / patient record database – A (1) / disk defragmenter – S (1) / device driver – S (1) / Google Chrome – A (1). 3. Any two differences, 2 marks each (point + contrast): packaged is made for many users, tailored for one organisation's needs (2) / packaged is cheaper because cost is shared, tailored is expensive (2) / packaged is available immediately, tailored takes time to develop (2) / packaged has lots of support / tutorials, tailored relies on the developer (2).